Lab Title: Modeling a University Course

Instructions: Read the following scenario carefully. You will create a single class to model this system. Your task is to decide what attributes and methods are necessary. **Make sure to test each step as you go.** You can only refer to course slides but not any information from the internet.

Scenario:
You're building a system for **North Park University** to manage its courses by creating a Course class. Each course must have:

- **Course code** (e.g., "CSIS1230")
- **Course title** (e.g., "Programming for Everyone")
- **Maximum student capacity**, max_students

Additionally, the university needs to **track how many courses** exist in a semester across all sections.

As students register, your system should:

1. Add them to the course roster.
2. Show course details clearly for users.
3. Provide a developer-friendly view for debugging.

Task 1: Blueprint the Class (10 mins)

- Define the Course class.
- Decide which attributes should be ***instance attributes*** (unique to each course) and which should be a ***class attribute*** (shared by all courses).
- Write the `__init__` method accordingly.

  Hint*: The scenario gives you clear hints for some instance attributes and one class attribute.*

Task 2: Add Core Functionality (10 mins)

Create an **instance method** that:

- Adds the student to the course's roster. (Tip: use a **list** to store the roster.)
- Checks if the course is already full (compare current roster size to max_students).

- If there's space, add the student and **print** a confirmation like:

  `Student Alice Smith has been added to CSIS1230.`

  Or

  `Cannot add student: course CSIS1230 is full.`

Task 3: Object Representation (10 mins)

- Implement **__str__** to provide a clean, user-friendly representation: e.g., "CSIS1230 - Programming for Everyone (25/30 students)"
- Implement the **__repr__** method. It should return a string that looks like a valid Python command to recreate the object, e.g.: "Course('CSIS1230', 'Programming for Everyone', 30)"

Task 4: Test Your Class (15 mins)

Write a script to test that your class works as expected:

- Create **two** Course instances - e.g., "CSIS1230" and " CSIS1210".
- Use **add_student** to try adding more students than max_students for at least one course, ensuring the full check works.
- Use both:
  - `print(course_instance)` to confirm __str__ output.
  - repr(course_instance) (wrapped in print()) to confirm __repr__ output.
- Print the **class attribute** tracking total courses to verify it correctly counts both objects.